

AAYUSH BHAT

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PROFESSIONAL SUMMARY

AI-integrated full-stack developer skilled in building end-to-end intelligent systems — from CNN-LSTM deep learning models and OpenAI API pipelines to React frontends and FastAPI backends. Developed **Mentora**, a real-time cognitive fatigue detection platform achieving a **0.93 Macro F1-Score**, and a production-grade Inventory Management System with JWT auth, RBAC, and Docker. Strong foundations in DSA, system design, DBMS, and cloud deployment on Vercel and Render.

EXPERIENCE

Software Engineering Intern

Jun 2023 – Aug 2023

IBM SkillsBuild (Virtual Internship)

Remote

- Designed and developed an AI-powered book recommendation chatbot using **IBM Watson Assistant** on IBM Cloud, enabling personalized user interactions via intent-based conversational workflows.
- Integrated genre, author, mood, and preference-based recommendation logic through structured dialog flows and entity recognition.
- Implemented API integration, asynchronous request handling, and dynamic UI updates using HTML, CSS, and JavaScript for a responsive, real-time user experience.

EDUCATION

Dr. APJ Abdul Kalam Technical University (AKTU)

2022 – 2026

Bachelor of Technology in Computer Science Engineering

Uttar Pradesh

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Redux, State Management, Responsive UI

Backend: FastAPI, Flask, Node.js, RESTful APIs, WebSocket APIs, JWT Authentication, API Integration

Database: PostgreSQL, Firebase (Firestore), Supabase, MongoDB

DevOps & Tools: Docker, Docker Compose, Git, GitHub, GitHub Actions, Postman, Pytest, CI/CD, Vercel, Netlify, Render

AI & Cloud: OpenAI APIs, Prompt Engineering, TensorFlow, CNN-LSTM, MediaPipe, OpenCV, IBM Watson, Google Cloud Platform (Firebase)

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Agile Methodology

PROJECTS

Mentora | *AI-Powered Cognitive Fatigue & Well-Being Tracker* | github.com/aayush1233333/mentora Jan 2025 – Mar 2025

- Developed a full-stack AI-powered fatigue monitoring platform for **real-time** cognitive fatigue detection using webcam-based facial analysis and deep learning.
- Trained and optimized a **CNN-LSTM** model on temporal facial landmark sequences, achieving a **0.93 Macro F1-Score** for multi-class fatigue classification across 3 fatigue states.
- Built scalable **REST APIs and WebSocket services** with FastAPI enabling real-time bidirectional predictions, session tracking, and sub-second frontend-backend communication.
- Integrated **MediaPipe, OpenCV, and TensorFlow** pipelines for facial landmark extraction, feature engineering, and live inference on streaming webcam data.
- Implemented secure authentication using **Firebase Auth, JWT, and Firestore** for user management, session history, and analytics storage.
- Integrated **OpenAI APIs** to deliver personalized wellness recommendations and fatigue-aware conversational support based on real-time session data.
- Deployed on **Render** (FastAPI backend) and **Vercel** (React frontend) with environment-based configuration and production-ready CI/CD workflows.

Inventory & Order Management System | github.com/aayush1233333/inventory-system

Jan 2026 – Mar 2026

- Developed a full-stack inventory and order management platform supporting end-to-end product, customer, and order lifecycle workflows across **6+ relational entities**.
- Built **15+ secure RESTful API endpoints** using FastAPI covering CRUD operations, inventory tracking, order processing, stock updates, validation, and structured error handling.

- Implemented **JWT Authentication and Role-Based Access Control (RBAC)** to enforce fine-grained user access and protect business-critical operations.
- Designed a normalized **PostgreSQL** schema across 6+ relational tables (products, orders, customers, inventory, users, roles) with indexing and transactional consistency, supporting concurrent multi-entity workflows.
- Automated inventory synchronization and order workflows, eliminating manual reconciliation for multi-SKU scenarios and ensuring accurate real-time stock management.
- Enhanced frontend performance through **React** state management and asynchronous API integration, delivering responsive user experiences with optimistic UI updates.
- Containerized the full application stack (API server, PostgreSQL, and supporting services) using **Docker and Docker Compose**, reducing environment setup time to under 5 minutes and ensuring dev/production parity.
- Tested and validated all REST API endpoints and system workflows using **Postman** and **Pytest** unit tests, ensuring robust error handling, data validation, and reliable endpoint performance.